PRACTICAL_6

Product Catalog Application using ASP.NET MVC

Subject: .NET

1. Aim

To create a Product Catalog mini-project using ASP.NET MVC architecture with Models, Views, Controllers, and Routing, and to provide options to view, add, edit, and delete products.

2. Technology Used

- ASP.NET MVC 5 (.NET Framework)
- C#
- Razor View Engine (.cshtml)
- Visual Studio

3. Pages Created

Page	File Location	Purpose (one sentence)
Product List	Views/Product/Index.cshtml	Displays all products with their ID, name, price, and links to add, edit, or delete products.
Add Product	Views/Product/Create.cshtml	Provides a form to enter a product name and price and submit a new product.
Edit Product	Views/Product/Edit.cshtml	Shows the selected product details and lets the user update its name and price.
Delete Product	Views/Product/Delete.cshtml	Shows the selected product details and asks the user to confirm before deleting it.

4. Code Files and Their Information

File	Location	Code information (one sentence)
Product.cs	Models/Product.cs	Defines the Product model with Id, Name, and Price properties.
ProductController.cs	Controllers/ProductController.cs	Handles displaying products and the Create, Edit, and Delete actions using a temporary in-memory list.
Index.cshtml	Views/Product/Index.cshtml	Displays the product catalog table and links to the Add, Edit, and Delete pages.
Create.cshtml	Views/Product/Create.cshtml	Contains the form used to enter and submit a new product.
Edit.cshtml	Views/Product/Edit.cshtml	Contains the form used to

		change the name and price of an existing product.
Delete.cshtml	Views/Product/Delete.cshtml	Displays product details and a confirmation form for deleting a product.
RouteConfig.cs	App_Start/RouteConfig.cs	Maps browser URLs to controller actions and sets Product/Index as the default route.
Global.asax.cs	Global.asax.cs	Starts the MVC application and registers the configured routes when the application starts.

5. How the Application Works

1. The browser requests the Product/Index route.
2. ProductController sends the product list to the Index view.
3. The Index page displays each product and provides Add, Edit, and Delete links.
4. Create adds a product, Edit updates a product, and Delete removes a product after confirmation.
5. After a successful change, the application redirects back to the Product List page.

6. Important Note

This practical stores products in a static in-memory list, not in a database; therefore, product changes are temporary and are lost when the application process restarts.

7. Conclusion

The Product Catalog mini-project was created using ASP.NET MVC and demonstrates the Model, View, Controller, and Routing concepts with product listing, adding, editing, and deleting.

8. Quick Viva Notes

Model: Represents the product data.

View: Displays the user interface.

Controller: Handles requests and connects the model with the view.

Routing: Maps a URL to a controller action.

Razor: Allows C# expressions to be used inside .cshtml views.